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Securely deliver applications with
Cloudflare

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Securely deliver applications with Cloudflare

Last reviewed: almost 2 years ago

Overview and the Cloudflare advantage

Cloudflare provides a complete suite of services around application performance, security, reliability, development, and Zero Trust. Cloudflare's global network is approximately 50 ms away from about 95% of the Internet-connected population and consists of

services that run on every server in every data center. The global scale of Cloudflare also allows for a robust threat intelligence source which is constantly fed back into Cloudflare security products to enhance the machine learning models and services even further.

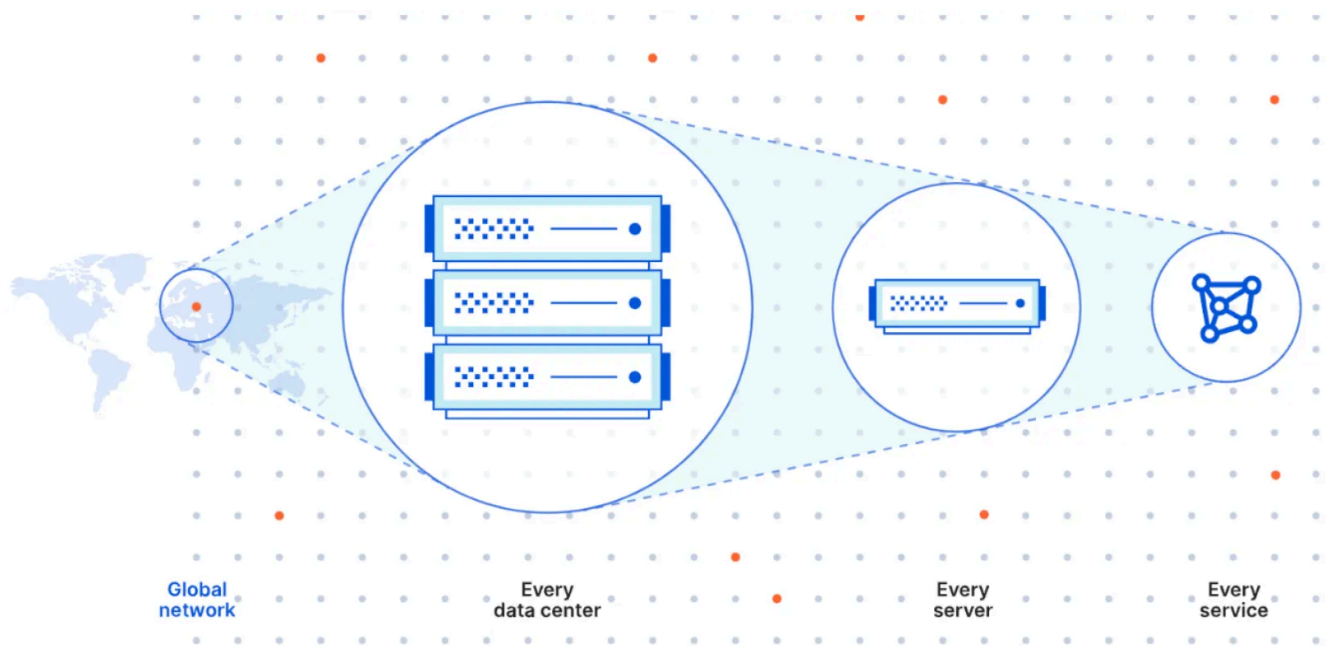


Figure 1: Cloudflare services run on every server in every data center

Other differentiators include the fact that Cloudflare is not a point product unlike some vendors who only offer API security or zero trust services or specific performance/security services. Customers have started moving away from the point-product approach due to operational and management complexities,

inefficiencies related to not being able to leverage cross-product innovation/integrations, and not being able to leverage scale of the network/resources across all services.

Integrated Global Cloud Platform

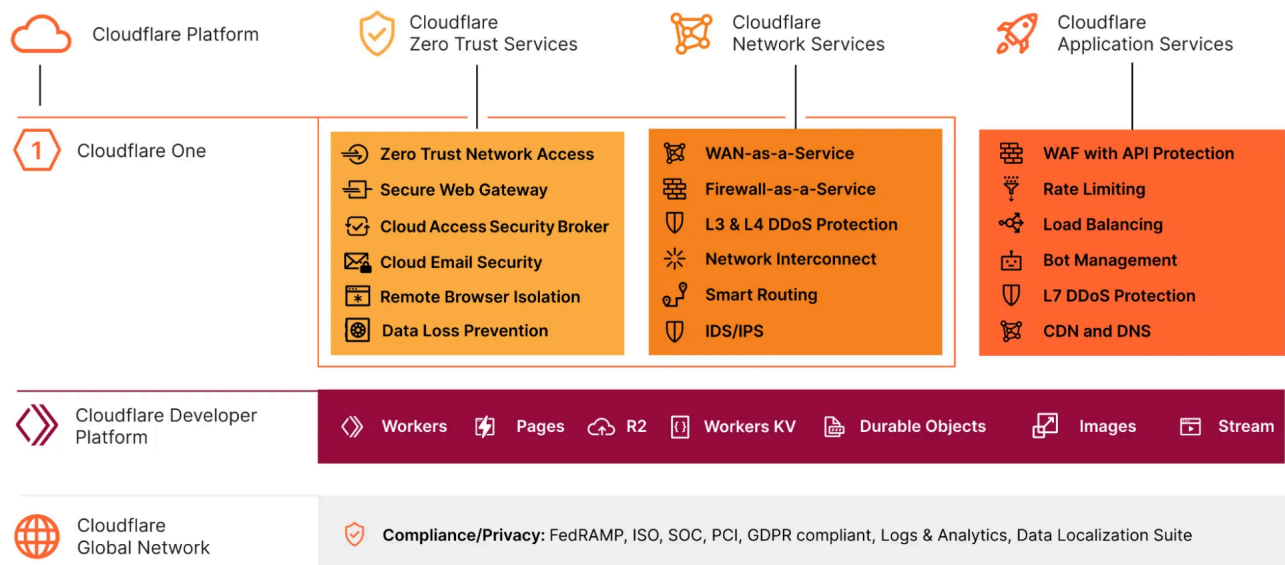


Figure 2: Cloudflare Global Cloud Platform.

Additionally, customers do not want to be locked in to a specific cloud provider, but many performance and security vendors lock customers into their platform by focusing on and optimizing services to their own cloud and making it operationally difficult to adopt a multi-cloud strategy.

Cloudflare is agnostic to where the workloads run or what cloud provider is being used. Customers get the same consistent unified dashboard and operational simplicity whether workloads run in a specific cloud or on-premise. Unlike many vendors, taking advantage of cross-product innovations and integration does not depend on customers using a specific cloud for workloads.

This document demonstrates how easy it is to use Cloudflare's collective services regardless of where workloads run. For the example in this document, an application workload will use Cloudflare DNS, CDN, WAF, and Access while also using Cloudflare Tunnel to connect securely to the Cloudflare network. It's rare for a vendor to provide this comprehensive level of security capability in an operationally simple and consistent fashion.

For additional details and reference architectures on specific services, see our [reference architecture documents](https://developers.cloudflare.com/reference-architecture/design-guides/secure-application-delivery/documents).

Onboarding and protecting the application with Cloudflare

Cloud-based security and performance providers like Cloudflare work as a reverse proxy. A reverse proxy is a server that sits in front of web servers and forwards client requests to those web servers. Reverse proxies are typically implemented to help increase security, performance, and reliability.

Normal traffic flow without a reverse proxy would involve a client sending a DNS lookup request, receiving the origin IP address, and communicating directly to the [origin server\(s\)](#).

When a reverse proxy is introduced, the client still sends a DNS lookup request to its resolver, which is the first stop in the DNS lookup. In some cases, the vendor providing the reverse proxy also provides DNS services; this is visualized in Figure 3 below. However, the client now communicates to the reverse proxy and the

reverse proxy communicates to the origin server(s). This traffic flow, where all traffic passes through the reverse proxy, allows for additional application security, performance, and reliability services to be implemented easily for applications.

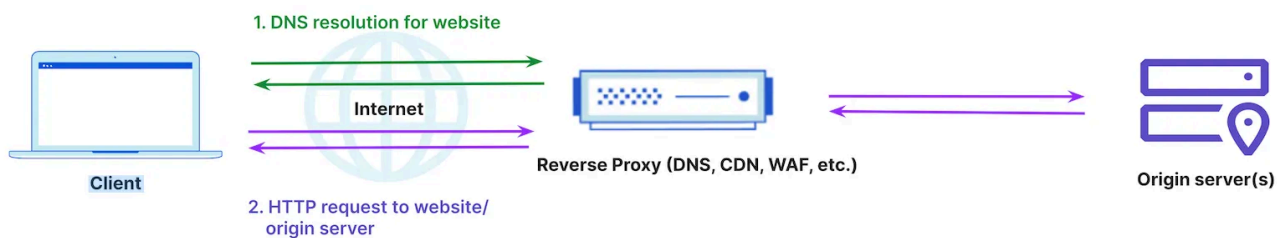


Figure 3: Same vendor providing DNS and security/performance services via proxy.

In this example, we have a website running on one of the major cloud providers and we want to use Cloudflare DNS, CDN, WAF, and Access. We want to start with these services for demonstration purposes; customers can expand these to include other Cloudflare services as desired. Cloudflare provides the benefit of decoupling all services from the cloud provider and if we want to change cloud providers later or protect other applications running in other

clouds, the dashboard and operations all stay consistent.

Customers can easily and securely connect their web application to the Cloudflare network and leverage application performance and security services. There are several connectivity options that fit different use cases.

Connectivity options

Public connection over the Internet

In the most basic scenario, the Cloudflare proxy will route the request traffic over the Internet to the origin. In this setup the client and origin are both endpoints directly connected to the Internet via their respective ISPs. The request is routed over the Internet from the client to Cloudflare proxy (via DNS configuration) before the proxy routes the request over the Internet to the customer's origin.

The below diagram describes the default connectivity to origins as requests flow through the Cloudflare network. When a request for the origin resolves to an IP hosted by Cloudflare, that request is then handled by the Cloudflare network and forwarded onto the origin server over the public Internet.

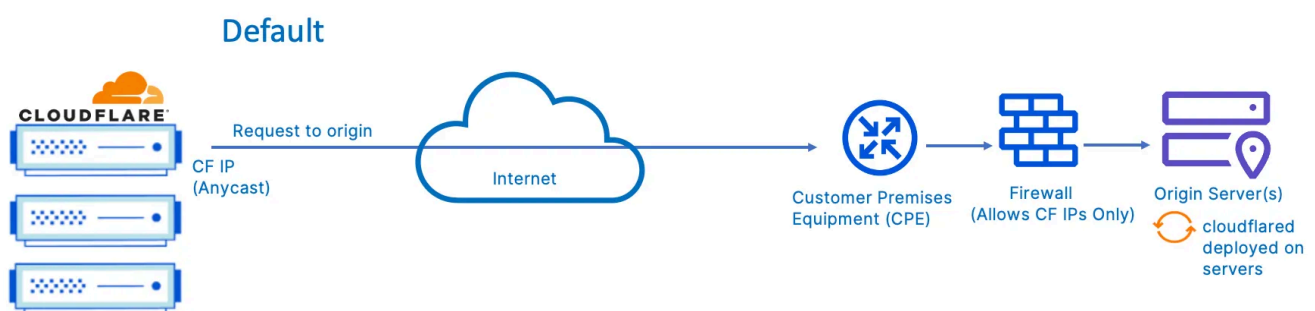


Figure 4: Connectivity from Cloudflare to origin server(s) via Internet

The origin is connected directly to the Internet and traffic is routed to the origin based on the IP address resolved by Cloudflare DNS. The DNS A record associates the domain name with the IP address of the origin server(s) or typically a load balancer the origin(s) are sitting behind.

In this model, when Cloudflare DNS receives a query for the A record, a Cloudflare anycast IP address is returned, so all traffic is routed through Cloudflare. However, unless additional precautions are taken, it's possible for the origin to be reached directly bypassing Cloudflare if someone knows the IP address of the origin(s).

Additionally, in this model, the customer has to open firewall rules for the origin(s) or web server(s) so they can be accessible on the respective http/https ports. However, customers can choose to leverage [Cloudflare Aegis](#), which allocates customer-specific IPs that Cloudflare will use to connect back to your origins. We recommend allowlisting traffic from only these networks to avoid direct access.

In addition to IP blocking at the origin-side firewall, we also strongly recommend additional verification of traffic via either the ["Full \(Strict\)" SSL setting](#) or [mTLS auth](#) to ensure all traffic is sourced from requests passing through the customer configured zones.

Cloudflare also supports [Bring Your Own IP \(BYOIP\)](#).

When BYOIP is configured, the Cloudflare global network will announce a customer's own IP prefixes and the prefixes can be used with the respective Cloudflare Layer 7 services. This allows customers to proxy traffic through Cloudflare and still have the customer IP address returned in the DNS resolution.

This can be [beneficial](#)  for cases where the customer IP prefixes are already allow-listed and updating firewall rules is not desirable or present an administrative hurdle.

Private connection over the Internet - Tunnel

The recommended option when connecting origin(s) over the Internet is to have a private tunnel/connection over the Internet for additional security.

A traditional VPN setup is not optimal due to backhauling traffic to a centralized VPN gateway location which then connects back to the origin; this negatively impacts end-to-end throughput and latency. Cloudflare offers [Cloudflare Tunnel](#) software that

provides an encrypted tunnel between your origin(s) and Cloudflare's network. Also, since Cloudflare leverages anycast on its global network, the origin(s) will, like clients, connect to the closest Cloudflare data center(s) and therefore optimize the end-to-end latency and throughput.

When you run a tunnel, a lightweight daemon in your infrastructure, `cloudflared`, establishes four outbound-only connections between the origin server and the Cloudflare network. These four connections are made to four different servers spread across at least two distinct data centers providing robust resiliency. It is possible to install many `cloudflared` instances to increase resilience between your origin servers and the Cloudflare network.

`Cloudflared` creates an encrypted tunnel between your origin web server(s) and Cloudflare's nearest data center(s), without the need for opening any public inbound ports. This provides for simplicity and speed of implementation as there are no security changes

needed on the firewall. This solution also lowers the risk of firewall misconfigurations which could leave your company vulnerable to attacks.

The firewall and security posture is hardened by locking down all origin server ports and protocols via your firewall. Once Cloudflare Tunnel is in place and respective security applied, all requests on HTTP/S ports are dropped, including volumetric DDoS attacks. Data breach attempts, such as snooping of data in transit or brute force login attacks, are blocked entirely.

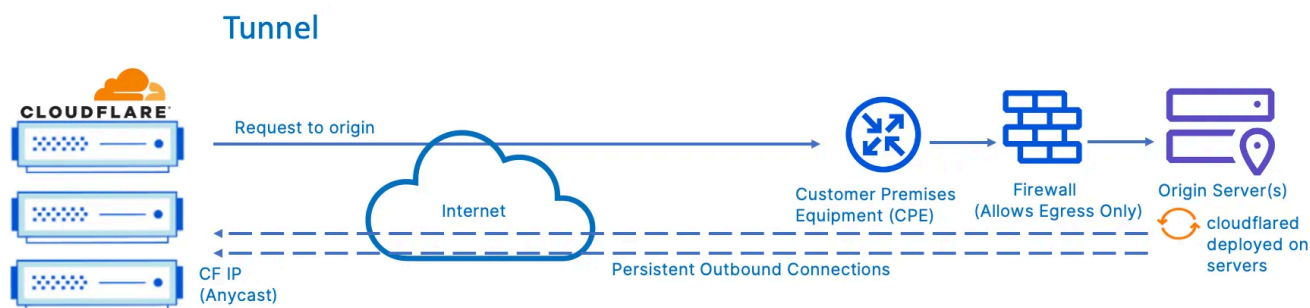


Figure 5: Connectivity from Cloudflare to origin server(s) via Cloudflare Tunnel

The above diagram describes the connectivity model through Cloudflare Tunnel. This option provides you with a secure way to connect your resources to Cloudflare without a publicly routable IP address.

Cloudflare Tunnel can connect HTTP web servers, SSH servers, remote desktops, and other protocols safely to Cloudflare.

Direct connection - Cloudflare Network Interconnect (CNI)

Most vendors also provide an option of directly connecting to their network. Direct connections provide security, reliability, and performance benefits over using the public Internet. These direct connections are done at peering facilities, Internet exchanges (IXs) where Internet service providers (ISPs) and Internet networks can interconnect with each other, or through vendor partners.

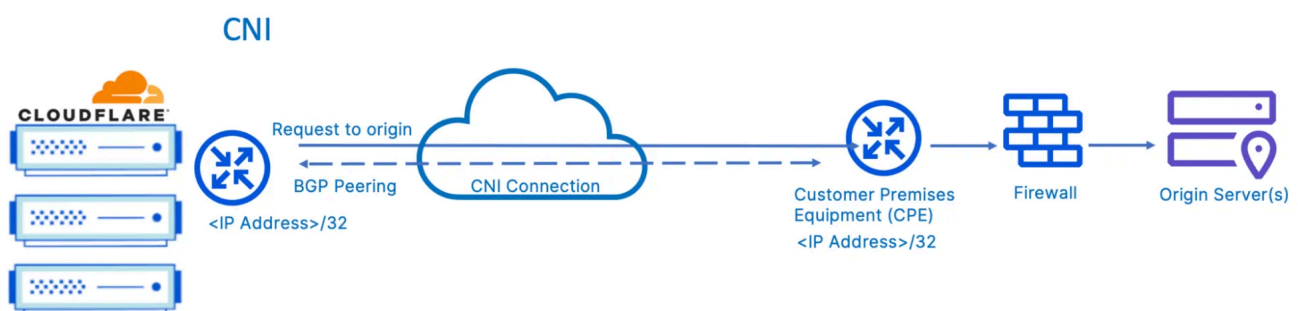


Figure 6: Connectivity from Cloudflare to origin server(s) via Cloudflare Network Interconnect (CNI)

The above diagram describes origin connectivity through [Cloudflare Network Interconnect \(CNI\)](#) ↗ which allows you to connect your network infrastructure directly with Cloudflare and communicate only over those direct links. CNI allows customers to interconnect branch and headquarter locations directly with Cloudflare. Customers can interconnect with Cloudflare in one of three ways: over a private network interconnect (PNI) available at [Cloudflare peering facilities](#) ↗, via an IX at any of the [many global exchanges Cloudflare participates in](#) ↗, or through one of Cloudflare's [interconnection platform partners](#) ↗.

Cloudflare's global network allows for ease of connecting to the network regardless of where your infrastructure and employees are.

Routing to the origin

Regardless of which connectivity model is used, DNS resolution is done first and provides Cloudflare the

information of where to route to. Cloudflare can support configurations as an authoritative DNS provider, secondary DNS provider, or non-Cloudflare DNS (CNAME) setups for a zone. For Cloudflare performance and security services to be applied, the traffic must be routed to the Cloudflare network.

Example: Securing your application with Cloudflare Tunnel and Access

Securing connectivity with Cloudflare Tunnel

Although there are multiple ways to onboard an application to use Cloudflare services, a common approach is to use Cloudflare DNS as the primary authoritative DNS. The additional benefit for customers here is that Cloudflare is consistently ranked the [fastest available authoritative DNS provider globally ↗](#).

In this example, we'll connect our origin server to Cloudflare securely with Cloudflare Tunnel. You can configure DNS in the dashboard and enter the site you

want to onboard. You'll receive a pair of Cloudflare nameservers to configure at your domain registrar's site. Once that's completed, Cloudflare becomes the primary authoritative DNS provider.

If Cloudflare is configured for just routing over the Internet, the DNS configuration would look something like below, where the A record points to the IP address of the origin server or respective load balancer. As Cloudflare is acting as a reverse proxy, the status shows as Proxied." As is, Cloudflare is still acting as a reverse proxy so all the Cloudflare services such as CDN, WAF, and Access can be used.

DNS

Records

Manage DNS records of your domain.

[DNS records documentation](#)

Required steps to complete zone set-up

[Hide](#)

- ✓ Add an A, AAAA, or CNAME record for **www** so that **www.cftestsite3.com** will resolve.
- ✓ Add an MX record for your **root domain** so that mail can reach @**cftestsite3.com** addresses or [set up restrictive SPF, DKIM, and DMARC records](#) to prevent email spoofing. [New Alert](#)

DNS management for **cftestsite3.com**

[Import and Export](#)[Dashboard Display Settings](#)

Review, add, and edit DNS records. Edits will go into effect once saved.

Search DNS Records

[Add filter](#)

Q

[Search](#)[Add record](#)

Type ▲	Name	Content	Proxy status	TTL	Actions
A	cftestsite3.com	54.193.122.124	 Proxied	Auto	Edit

Figure 7: DNS configuration for 'cftestsite3.com' - pointing to IP address of origin or load balancer.

We can also use Cloudflare Tunnel over the Internet to provide for more security and to prevent the need for opening any inbound firewall rules to the origin(s). In this way, instead of an A record in the DNS configuration, we will have a CNAME record pointing to the tunnel we deploy. Here we deploy a tunnel from the origin to the Cloudflare network, and the DNS will automatically be configured. A CNAME record that points to the tunnel will be created; this enforces all traffic going to the origin(s) be routed over the Cloudflare Tunnel.

To create and manage tunnels, you need to install and authenticate cloudflared on your origin server.

cloudflared is what connects your server to Cloudflare's global network.

There are two options for creating a tunnel - [via the dashboard](#) or [via the command line](#). It's recommended getting started with the dashboard, since it will allow you to manage the tunnel from any machine.

A remotely-managed tunnel only requires the tunnel token to run. Anyone with access to the token will be able to run the tunnel. You can get a tunnel's token from the dashboard or via the API as shown below. The command provided in the dashboard will install and configure cloudflared to run as a service using an auth token.

In the Cloudflare dashboard, navigate to Zero Trust > Access > Tunnels. Select the "Create a tunnel" button, name the tunnel, and save.

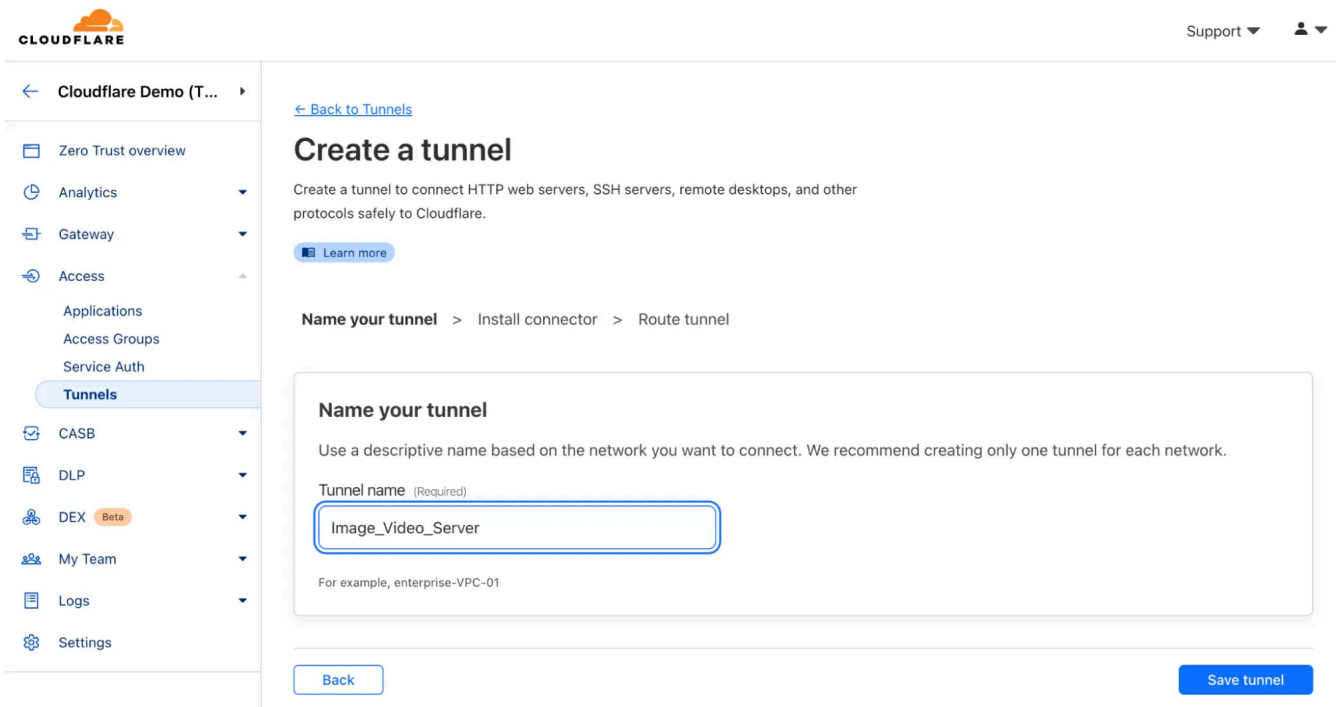


Figure 8: Cloudflare Tunnel Creation.

Next, you'll be presented with a screen where you select the operating system (OS) of your origin server. You will then be provided a CLI command that you can run on your origin that will automatically download and install the Cloudflare Tunnel software.

Cloudflare Demo (T...)

Zero Trust overview

Analytics

Gateway

Access

Applications

Access Groups

Service Auth

Tunnels

CASB

DLP

DEX Beta

My Team

Logs

Settings

[Back to Tunnels](#)

Configure Image_Video_Server

Name your tunnel > **Install connector** > Route tunnel

Choose your environment

Choose an operating system:

Windows Mac Debian Red Hat Docker

Install and run a connector

To connect your tunnel to Cloudflare, copy-paste one of the following commands into a terminal window. Remotely managed tunnels require that you install cloudflared 2022.03.04 or later.

Store your token carefully. This command includes a sensitive token that allows the connector to run. Anyone with access to this token will be able to run the tunnel.

If you don't have cloudflared installed on your machine:

```
brew install cloudflare/cloudflare/cloudflared && sudo cloudflared service install
```

If you already have cloudflared installed on your machine:

```
sudo cloudflared service install
```

[View Frequently Asked Questions](#)

Connectors

Your connectors will automatically show here once cloudflared has been successfully installed on your machine.

No connectors installed

Figure 9: Instructions to install and run a connector.

Below, the CLI command has been run to download and install the Cloudflare Tunnel software.


```

bitnami@ip-10-3-0-133:~$ curl -L --output cloudflared.deb https://github.com/cloudflare/cloudflared/releases/latest/download/cloudflared-linux-amd64.deb &&

sudo dpkg -i cloudflared.deb &&

sudo cloudflared service install

```

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
			Dload	Upload	Total	Spent	Left
0	0	0	0	0	--:--:--	--:--:--	0
0	0	0	0	0	--:--:--	--:--:--	0
100	100	100	26.6M	0	--:--:--	--:--:--	72.5M

```

Selecting previously unselected package cloudflared.
(Reading database ... 38410 files and directories currently installed.)
Preparing to unpack cloudflared.deb ...
Unpacking cloudflared (2023.8.2) ...
Setting up cloudflared (2023.8.2) ...
Processing triggers for man-db (2.9.4-2) ...
2023-09-05T19:02:02Z INF Using Systemd
2023-09-05T19:02:03Z INF Linux service for cloudflared installed successfully
bitnami@ip-10-3-0-133:~$

```

Figure 10: Downloading and installing Cloudflare Tunnel

The connector will now automatically be displayed as connected.

Install and run a connector

To connect your tunnel to Cloudflare, copy-paste one of the following commands into a terminal window. Remotely managed tunnels require that you install cloudflared 2022.03.04 or later.

Store your token carefully. This command includes a sensitive token that allows the connector to run. Anyone with access to this token will be able to run the tunnel. ✕

If you don't have cloudflared installed on your machine:

```
curl -L --output cloudflared.deb https://github.com/cloudflare/cloudflared/releases/latest/download/cloudflared-linux-amd64.deb &&
sudo dpkg -i cloudflared.deb &&
```

If you already have cloudflared installed on your machine:

```
sudo cloudflared service install
```

[View Frequently Asked Questions](#)

Connectors

Connector ID	Status	Version
78a7e5bc-6091-4bfc-b7e4-[REDACTED]	Connected	2023.8.2

Back

Next

Figure 11: Cloudflare Tunnel Connectors showing in dashboard.

In the dashboard, you can now continue with the next step which is to create the tunnel and map it to a service on the origin as shown below. In this case, all HTTPS traffic will be sent over the tunnel to the origin server.

The screenshot shows the Cloudflare dashboard with the 'Tunnels' section selected in the left sidebar. The main content area is titled 'Route Traffic for Image and Video Gallery'. It includes a breadcrumb trail: '← Back to Tunnels > Name your tunnel > Install connector > Route tunnel'. Below this, there are two informational boxes: one about routing traffic by adding a public hostname or private network, and another about protecting resources with an Access policy. The 'Public Hostnames' tab is active, showing a form to 'Add public hostname for Image and Video Gallery'. The form fields are: 'Subdomain' (optional, with placeholder '(optional) subdomain'), 'Domain' (required, dropdown menu showing 'cftestsite3.com'), and 'Path' (optional, with placeholder '(optional) path'). Below these is the 'Service' section with 'Type' (required, dropdown menu showing 'HTTPS') and 'URL' (required, text input showing 'localhost:443'). A note below the URL says 'For example, https://localhost:8001'. At the bottom of the form is a link 'Additional application settings >'. Navigation buttons 'Back' and 'Save tunnel' are at the bottom of the page.

Figure 12: Cloudflare Tunnel Configuration.

You can now see in the dashboard that the tunnel has been created and is healthy.

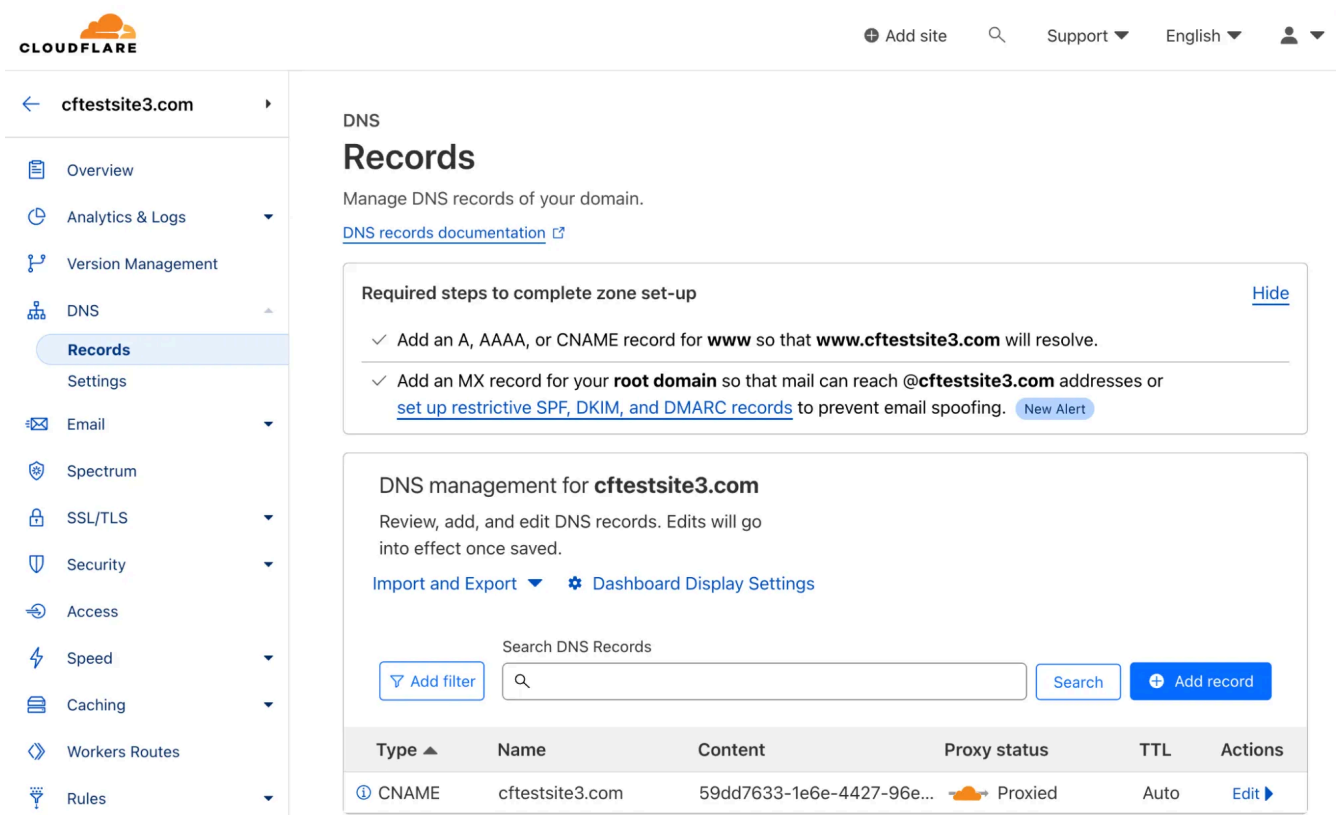
The screenshot shows the Cloudflare dashboard with the 'Tunnels' section selected in the left sidebar. The main content area is titled 'Tunnels' and includes a sub-header 'Tunnels establish a secure connection between Cloudflare's edge and your infrastructure.' Below this is a 'Learn more' link. A yellow banner message states: 'A new version of cloudflared is available. Visit our [downloads](#) page to upgrade.' Below the banner is a section titled 'Your tunnels' with a sub-header 'Showing 1 - 1' and a description 'Manage the configurations of your existing tunnels.' There is a '+ Create a tunnel' button and a search bar containing 'Image and Video Gallery'. A table lists the tunnels:

Tunnel name ↑	Tunnel ID	Status	Routes	Uptime	Created
Image and Video Gallery	59dd7633-1e6e-4427-96e3- [REDACTED]	HEALTHY	cftestsite3.com	38 days	September 5, 2023

At the bottom of the table, there is a pagination bar showing '1 - 1' and 'Items per page: 10'. On the right side of the pagination bar, there is a link '< 1 of 1 page >'.

Figure 13: Cloudflare Tunnel is created and healthy.

Further, if we look at the DNS configuration, we can see a DNS record was automatically created pointing to the tunnel ID. When you create a tunnel, Cloudflare generates a subdomain of `cfargotunnel.com` with the UUID of the created tunnel. Unlike publicly routable IP addresses, the subdomain will only proxy traffic for a DNS record in the same Cloudflare account. It's not possible for another user to create a DNS record in another account or system to proxy traffic over this tunnel.



The screenshot displays the Cloudflare dashboard for the domain `cftestsite3.com`. The left sidebar shows the navigation menu with 'DNS' selected. The main content area is titled 'DNS Records' and includes a 'Required steps to complete zone set-up' section with two checklist items: adding an A, AAAA, or CNAME record for `www` and adding an MX record for the root domain. Below this, the 'DNS management for cftestsite3.com' section shows a table of DNS records. The table has columns for Type, Name, Content, Proxy status, TTL, and Actions. A single CNAME record is listed for `cftestsite3.com`, with its content being a long alphanumeric string (the tunnel ID). The proxy status is 'Proxied' (indicated by a cloud icon) and the TTL is 'Auto'.

Type	Name	Content	Proxy status	TTL	Actions
CNAME	cftestsite3.com	59dd7633-1e6e-4427-96e...	Proxied	Auto	Edit

Figure 14: Cloudflare DNS CNAME record automatically created

We now have secure application access. Users can only access the application through the tunnel connected to the Cloudflare network. Further, since Tunnel uses outbound connections to Cloudflare and any return traffic from an outbound connection will be allowed, no inbound firewall rule is required creating less overhead and more operational simplicity.

If you were to deploy the tunnel via CLI, after the tunnel install, you would also need to authenticate [cloudflared](#) on the origin server. cloudflared is what connects the server to Cloudflare's global network. This authentication can be done with the `cloudflared tunnel login` command as shown below.

```
bitnami@ip-10-3-0-133:~$ cloudflared tunnel login
Please open the following URL and log in with your Cloudflare account:
```

```
https://dash.cloudflare.com/argotunnel?aud=&callback=https%3A%2F%2Flogin.cloudflareaccess.org%[REDACTED]
```

```
Leave cloudflared running to download the cert automatically.
```

Figure 15: Authenticating cloudflared on the origin server.

You'll be asked to select the zone you want to add the tunnel to as shown below.



Support ▾

English ▾

**Authorize Cloudflare Tunnel**

Cloudflare Tunnel wishes to serve as an origin on one of your zones.

Please select the zone you want to add a Tunnel to.

Search across all accounts...

cftestsite3.com

Search

Figure 16: Adding Cloudflare Tunnel to a selected zone.

Next, you'll authorize the tunnel for the zone.

Authorize Tunnel for cftestsite3.com

To finish configuring Tunnel for your zone, click Authorize below.

Cancel

Authorize

Figure 17: Authorizing the tunnel for a zone.

Finally, you should receive confirmation that a certificate has been installed allowing your origin to create a tunnel on the respective zone.

Success

Cloudflared has installed a certificate allowing your origin to create a Tunnel on this zone.

You may now close this window and start your Cloudflare Tunnel!

Figure 18: Confirmation that certificate has been successfully installed.

Securing the application with Cloudflare Access

The current setup as described prior in this document is shown below, where the origin server(s) are connected to the Cloudflare network via Tunnel. Now, we can start to consume Cloudflare services.

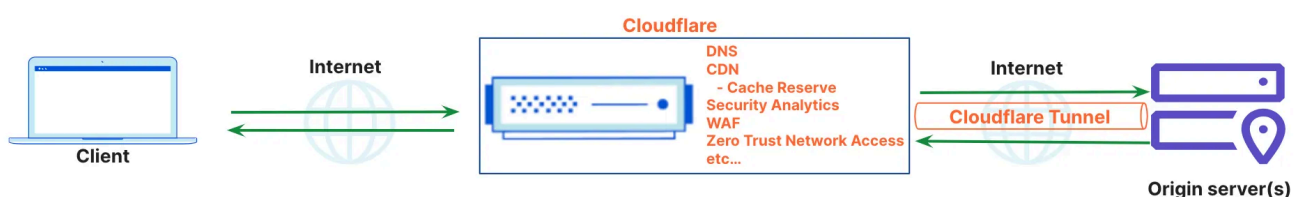


Figure 19: Web app securely connected to Cloudflare network for performance and security services.

Currently the origin is only accessible via Cloudflare Tunnel. Because a public hostname is used, access to

the origin is public. The application is secured behind Cloudflare and protected from DDoS and other types of attacks. For additional security, Cloudflare Access can be used to place a layer of authentication and access controls in front of the tunneled application. Access enforces an authentication step before requests to the origin can be served. Many other identity, device and network attributes can be used in the policy, allowing customers to define access beyond just authentication. For example, customers can define the network the request originates from, as well as ensuring the user device is running the latest operating system.

Below, you can see an application has been created for `cftestsite3.com`.

The screenshot shows the Cloudflare dashboard with the 'Applications' section selected in the left sidebar. The main content area is titled 'Applications' and includes a description: 'Protect your Self-Hosted, SaaS and Private applications with Zero Trust policies. Only users who match your policies will have access to your configured applications.' Below this, there's a section 'Your applications' showing a table with one application: 'Image and Video Gallery' with URL 'cftestsite3.com', 1 total domain, 1 policy assigned, and type 'SELF-HOSTED'. A search bar at the top right contains 'cftestsite3.com'.

Application name	Application URL	Total Domains	Policies assigned	Type
Image and Video Gallery	cftestsite3.com	1	1	SELF-HOSTED

Figure 20: Cloudflare Access Policy Configuration.

Looking at policy configuration below you can see it requires users to be part of the "Secure Employees" Access group.

The screenshot shows the 'Authorized Gallery Users Policy' configuration page. It includes fields for 'Policy name' (Authorized Gallery Users Policy), 'Action' (Allow), and 'Session duration' (Same as application session timeout). Below this is a section 'Assign a group' showing a list of access groups. The 'Secure Employees' group is selected and assigned to the policy.

Name	Action
☒ Secure Employees	Include
☐ Everyone	Include
☐ Azure Demo	Include
☐ Contractors	Include
☐ Devices on latest operating system	Include

Figure 21 : Access group assigned to the application.

If we take a deeper look at the "Secure Employees" Access group, it can be seen below that members are from the company's Okta identity provider (IdP) group called "Employees." Further, the Access group is enforcing multi-factor authentication (MFA).

The screenshot shows the Cloudflare Access Groups configuration interface. On the left is a sidebar with navigation links: Zero Trust overview, Analytics, Gateway, Access, Applications, Access Groups (selected), Service Auth, Tags (Beta), Tunnels, CASB, DLP, DEX (Beta), My Team, Logs, and Settings. The main content area is titled 'Secure Employees' and includes a 'Back to Groups' link. Below the title is a description: 'Create a new rule group to use across multiple Access applications. With groups, you can grant or restrict user access to your applications.' The 'Group name' field is 'Secure Employees' and the 'Group ID' is 'd21ce80f-5a25-4bcf-9b07-5032197e3be8' with a 'Copy' button. There is a section 'Assign as default group' with a checkbox 'Set as default group' which is currently unchecked. Below this is the 'Define group criteria' section, which instructs to 'Build rules to determine membership of this group. Your group must contain at least one Include rule.' It contains two rules: an 'Include' rule with 'Okta Groups' as the selector and 'Employees' as the value, and a 'Require' rule with 'Authentication Method' as the selector and 'mfa - multiple-factor authentication' as the value. At the bottom of the criteria section are links to '+ Add include', '+ Add require', and '+ Add exclude'.

Figure 22 : Access group configuration with defined group criteria.

Looking at the "Image and Video Gallery" application, under "Authentication," customers can also manually

select identity providers users can use to connect to this application.

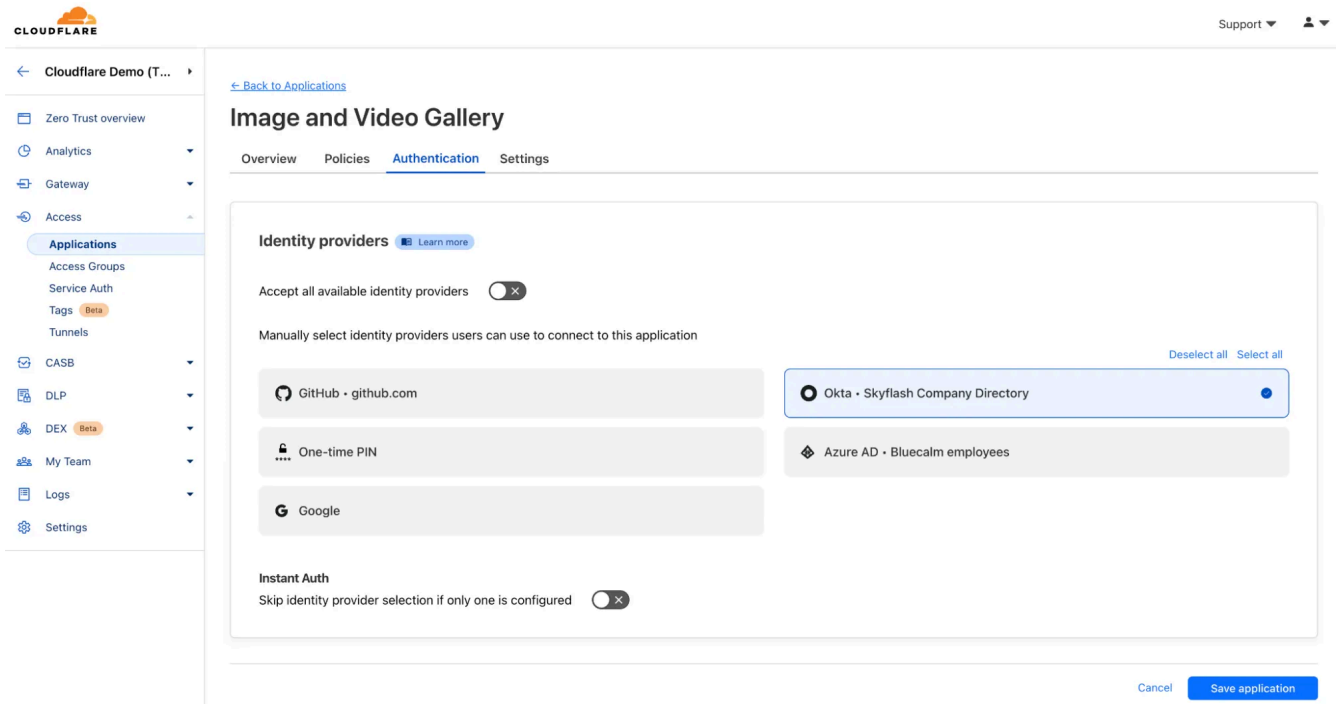


Figure 23 : Manually selecting identity providers users can use.

We now have secure application access to the origin(s) via Tunnel and also authentication and access policies to the application via Access. When users try to access the site, they are greeted with a Cloudflare Access page asking users to authenticate with the configured IdP; the page can be customized to customer's liking as shown below.

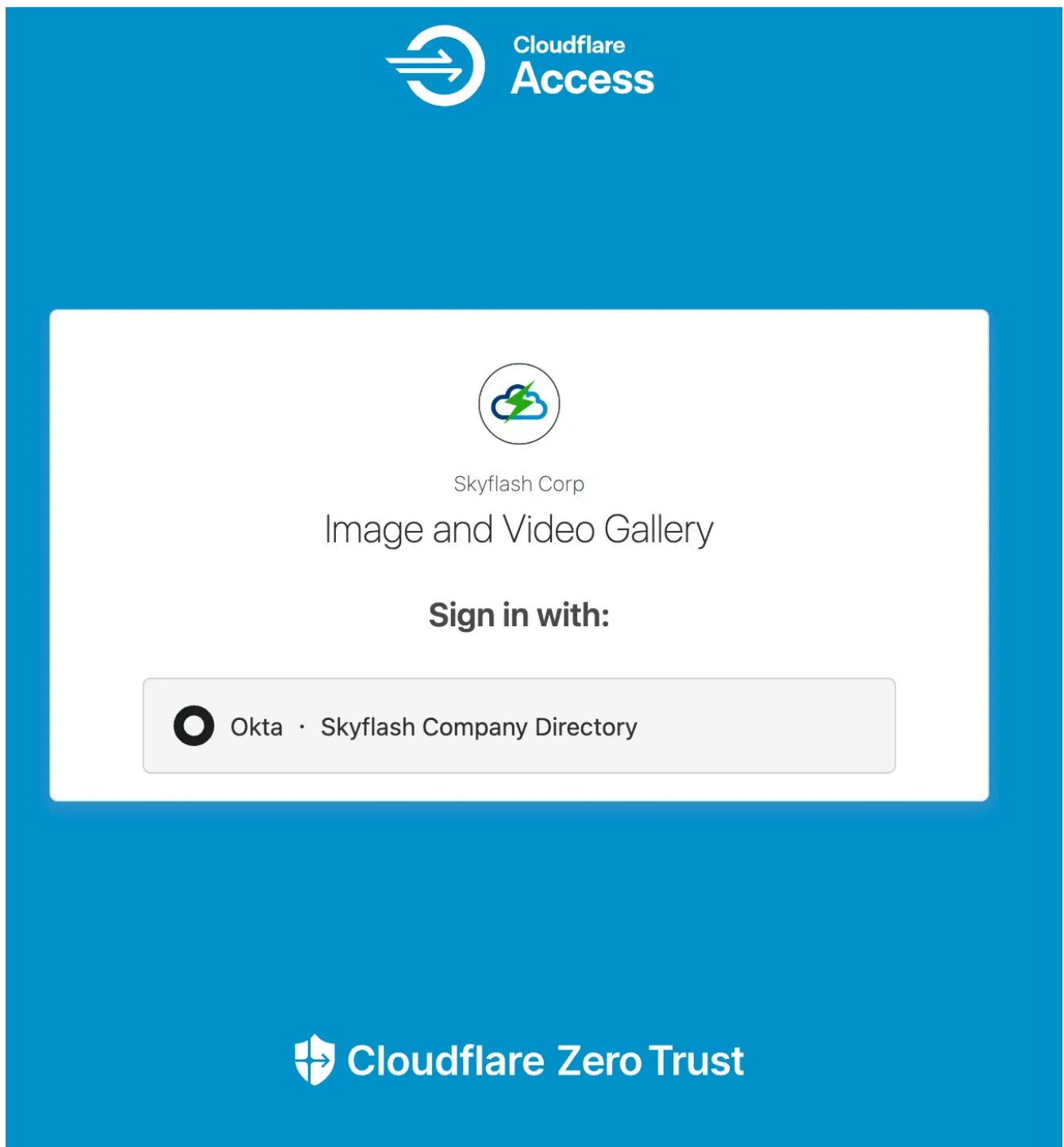


Figure 24 : Sign-in via IdP configured in Access.

Using other Cloudflare services (CDN, WAF, Security Analytics, etc.)

In the current setup, the origin server(s) are securely connected to the Cloudflare network via Cloudflare Tunnel and Cloudflare Access via policies enforcing authentication and other security requirements.

Since Cloudflare is already set up and acting as a reverse proxy for the site, traffic is being directed through Cloudflare, so all Cloudflare services can easily be leveraged including CDN, Security Analytics, WAF, API Shield, Bot Management, Page Shield for client-side security, etc.

When a DNS lookup request is made by a client for the respective website, in this case "cftestsite3.com," Cloudflare returns an anycast IP address, so all traffic is directed to the closest data center where all services will be applied before the request is forwarded over Cloudflare Tunnel to the origin server(s).

Cloudflare CDN leverages Cloudflare's global anycast edge network. In addition to using anycast for network performance and resiliency, the Cloudflare CDN leverages [Argo Tiered Cache](#) to deliver optimized

results while saving costs for customers. Customers can also enable [Argo Smart Routing](#) to find the fastest network path to route requests to the origin server. As shown below, the Cloudflare CDN is now caching content globally and granular CDN policies to affect default behavior can be applied.

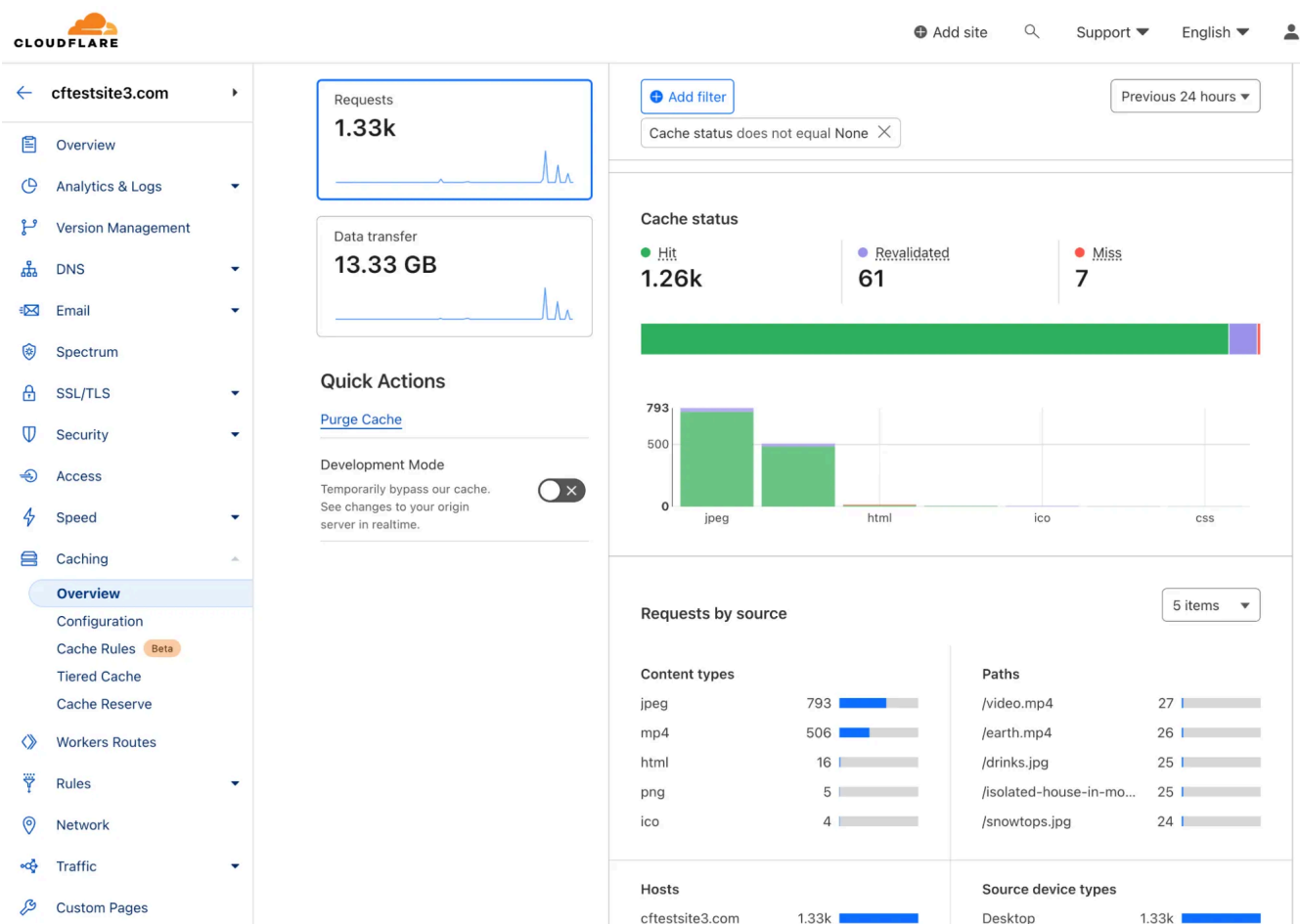


Figure 25 : Cloudflare Caching Analytics.

There are [different caching topologies and configurations available](#). Below, you can see a Cache

Rule has been configured to cache requests to the domain and override the origin TTL.

Cloudflare

← cftestsite3.com

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Create new Cache Rule

Rule name (required)
Override origin - 12 hr TTL
Give your rule a descriptive name.

When incoming requests match...

Field	Operator	Value
Hostname	equals	cftestsite3.com e.g. example.com

And Or

Expression Preview
(http.host eq "cftestsite3.com")
[Edit expression](#)

Then...

Cache eligibility (required)
Mark whether the request's response from origin is eligible for caching. Caching itself will still depend on the cache-control header and your other caching configurations.

☐ Bypass cache
☒ **Eligible for cache**

Edge TTL (optional)
Specify if and how long Cloudflare should cache the response, depending on if a cache-control header is present on the origin response.

☐ Use cache-control header if present, bypass cache if not
☐ Use cache-control header if present, use default Cloudflare caching behavior if not
☒ **Ignore cache-control header and use this TTL**

Input time-to-live (TTL) (required)
12 hours

Status code TTL
Specify how long Cloudflare should cache the response based on the status code from the origin.
[+ Add status code setting](#)

Figure 26 : Cloudflare rule configuration.

[Cloudflare Cache Reserve](#) has also been enabled by clicking the "Enable storage sync" button under "Caching > Cache Reserve" in the dashboard. Cache Reserve leverages Cloudflare's persistent object storage, R2, to eliminate egress costs from other public

cloud providers. It improves cache hit ratios by enabling customers to persistently cache data with the push of a single button.

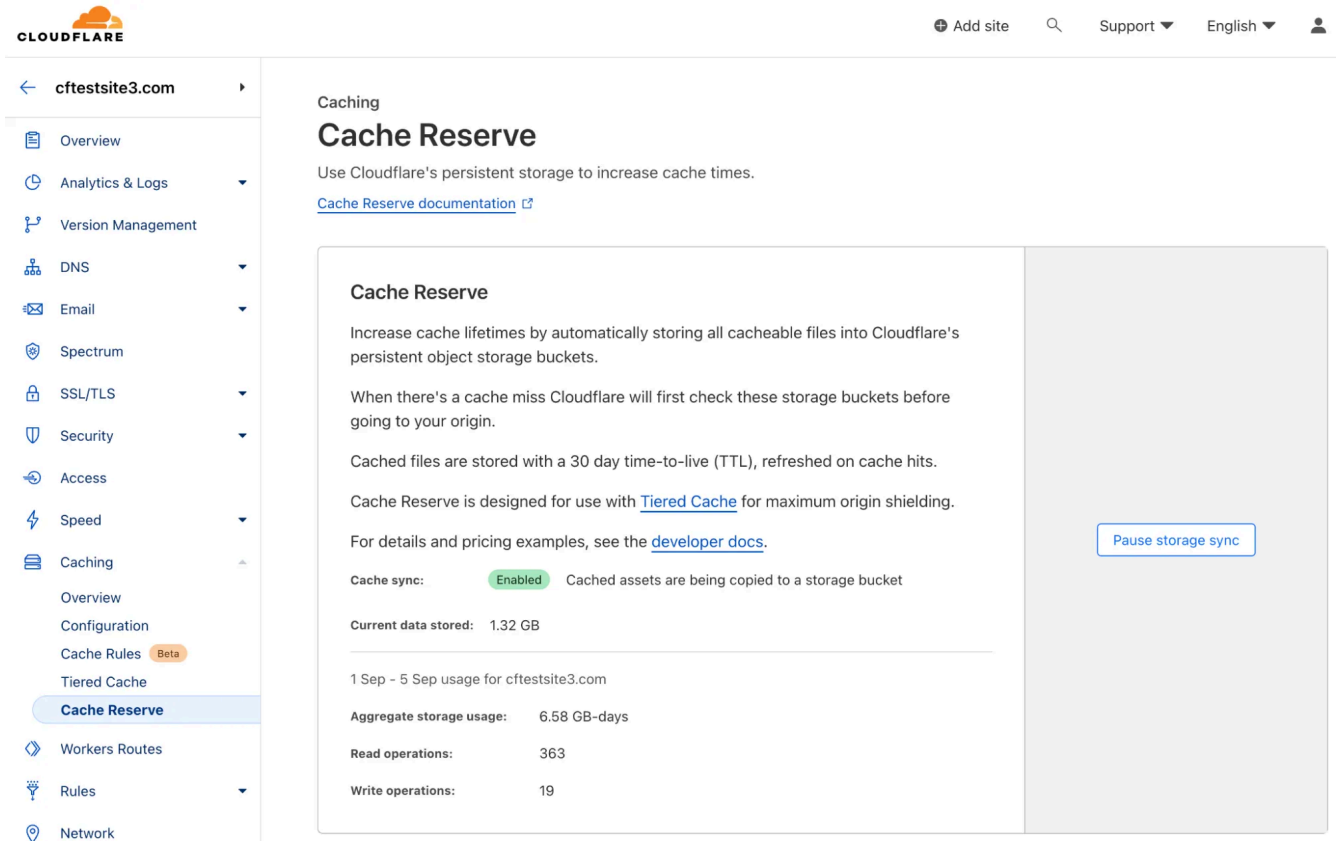


Figure 27 : Cloudflare Cache Reserve.

Additionally, as shown below, Cloudflare Security Analytics brings together all of Cloudflare's detection capabilities and provides a global view and important insights for all traffic going to the respective site. As traffic is being routed through the Cloudflare network, Cloudflare has visibility into threats and insights which

are exposed to customers in the dashboard, logs, and reporting.

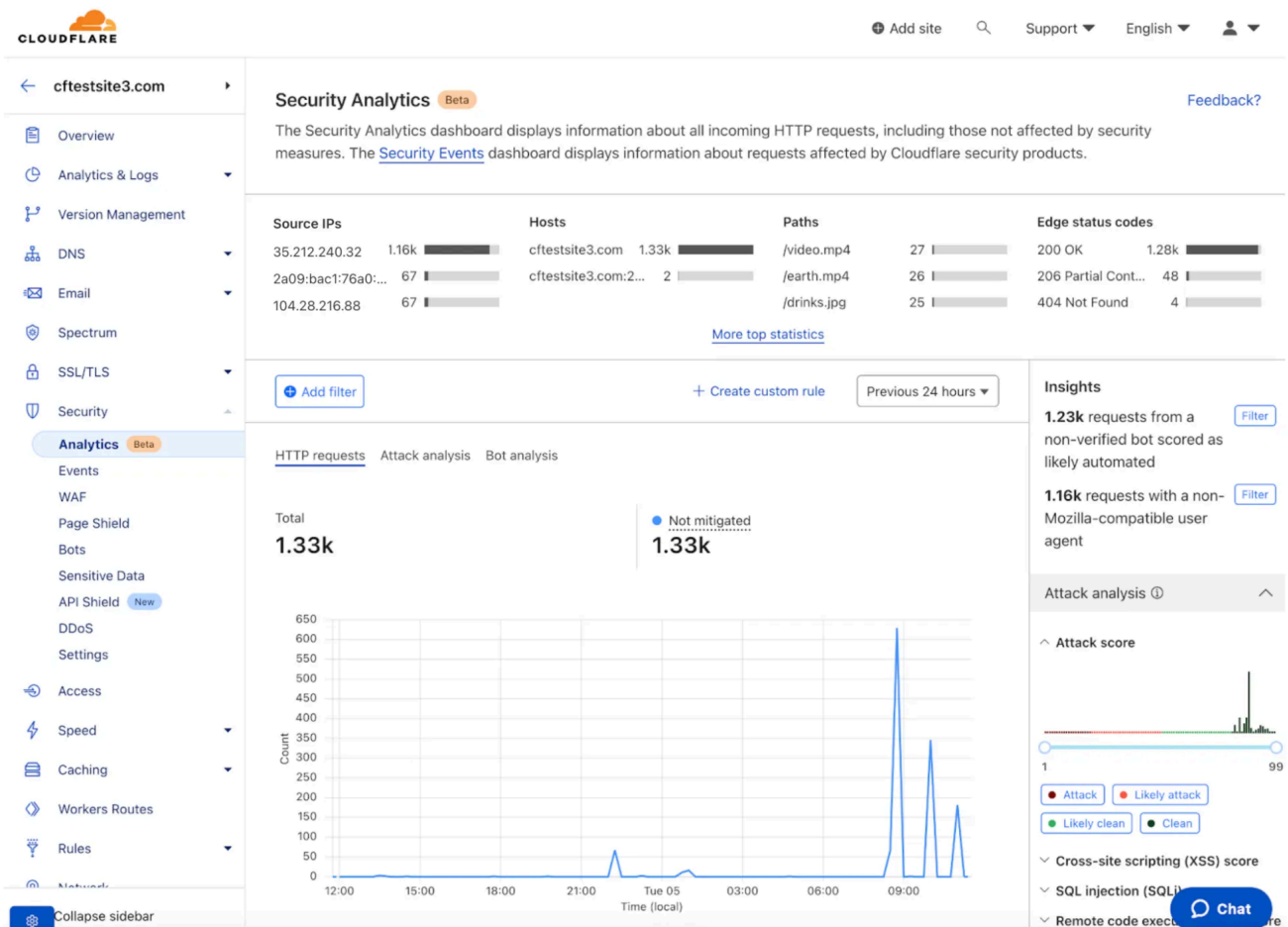


Figure 28 : Cloudflare Security Analytics.

Cloudflare WAF rules can be applied to enforce policies on traffic inline. Below a firewall policy is in place to log all traffic with a bot score of < 30 and WAF attack score < 50 . A bot score of < 30 signifies all traffic that's classified as either automated or likely automated and a WAF attack score < 50 signifies all traffic that's classified as either malicious or likely malicious.

Cloudflare

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Overview

Analytics & Logs

Version Management

DNS

Email

Spectrum

SSL/TLS

Security

Analytics Beta

Events

WAF

Page Shield

Bots

Sensitive Data

API Shield New

Security WAF

Zone-level Web Application Firewall (WAF) detects and mitigates malicious requests across all traffic under this zone.

[WAF documentation](#)

Custom rules Rate limiting rules Managed rules Update Tools

Protect your website and API from malicious traffic with custom rules. Configure mitigation criteria and actions, or [explore templates](#), for better security.

You have used **1 out of 1000** available rules.

[+ Create rule](#)

Search... Search Show filters

Order	Action	Name	CSR ⓘ	Activity last 24hr	Enabled
1	Log	Bot Score < 30 and WAF Attack Score < 50 Bot Score, WAF Attack Score	-	6.79k	✓

Figure 29 : Cloudflare WAF.

Cloudflare WAF allows for granular policies that can leverage many different request criteria including header information. Customers can take a [variety of actions](#) including logging, blocking, and challenge.

Security

WAF

Zone-level Web Application Firewall (WAF) detects and mitigates malicious requests across all traffic under this zone.

[WAF documentation](#)

Custom rules Rate limiting rules Managed rules Update Tools

[← Back](#)

Create rule [About custom rules](#)

Rule name (required)

Bot Score < 30 and WAF Attack Score < 50

Give your rule a descriptive name.

If incoming requests match...

Field	Operator	Value	
Bot Score	less than	30	And
e.g. 5			
And			
WAF Attack S...	less than	50	And Or
e.g. 5			

Expression Preview [Edit expression](#)

```
(cf.bot_management.score lt 30 and cf.waf.score lt 50)
```

Then take action...

Choose action

Log

Managed Challenge

Log

Block

JS Challenge

Skip

Interactive Challenge

Cancel

Save as Draft

Deploy

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Figure 30 : Cloudflare WAF Rule Configuration.

Customers can use WAF to implement and use custom rules, rate limiting rules, and managed rules. A brief description of each is provided below.

- WAF Custom Rules: provides ability to create custom rules based on different request attributes and header information to block any threat
- WAF Rate Limiting Rules: prevents abuse, DDoS, brute force attempts, and provides for API-centric controls.
- WAF Managed Rules
 - Cloudflare Managed Ruleset: provides advanced zero-day vulnerability protection
 - Cloudflare OWASP Core Ruleset: block common web application vulnerabilities, some of which are in OWASP top 10
 - Cloudflare Leaked Credential Check: checks exposed credential database for popular content management system (CMS) applications

The same methodology applies for all other Cloudflare Application Performance and Security products (API Shield, Bot Management, etc.): once configured to

route traffic through the Cloudflare network, customers can start leveraging the Cloudflare services. Figure 31 displays Cloudflare's Bot Analytics which categorizes the traffic based on bot score, shows the bot score distribution, and other bot analytics. All of the request data is captured inline and all enforcement based on defined policies is also done inline.

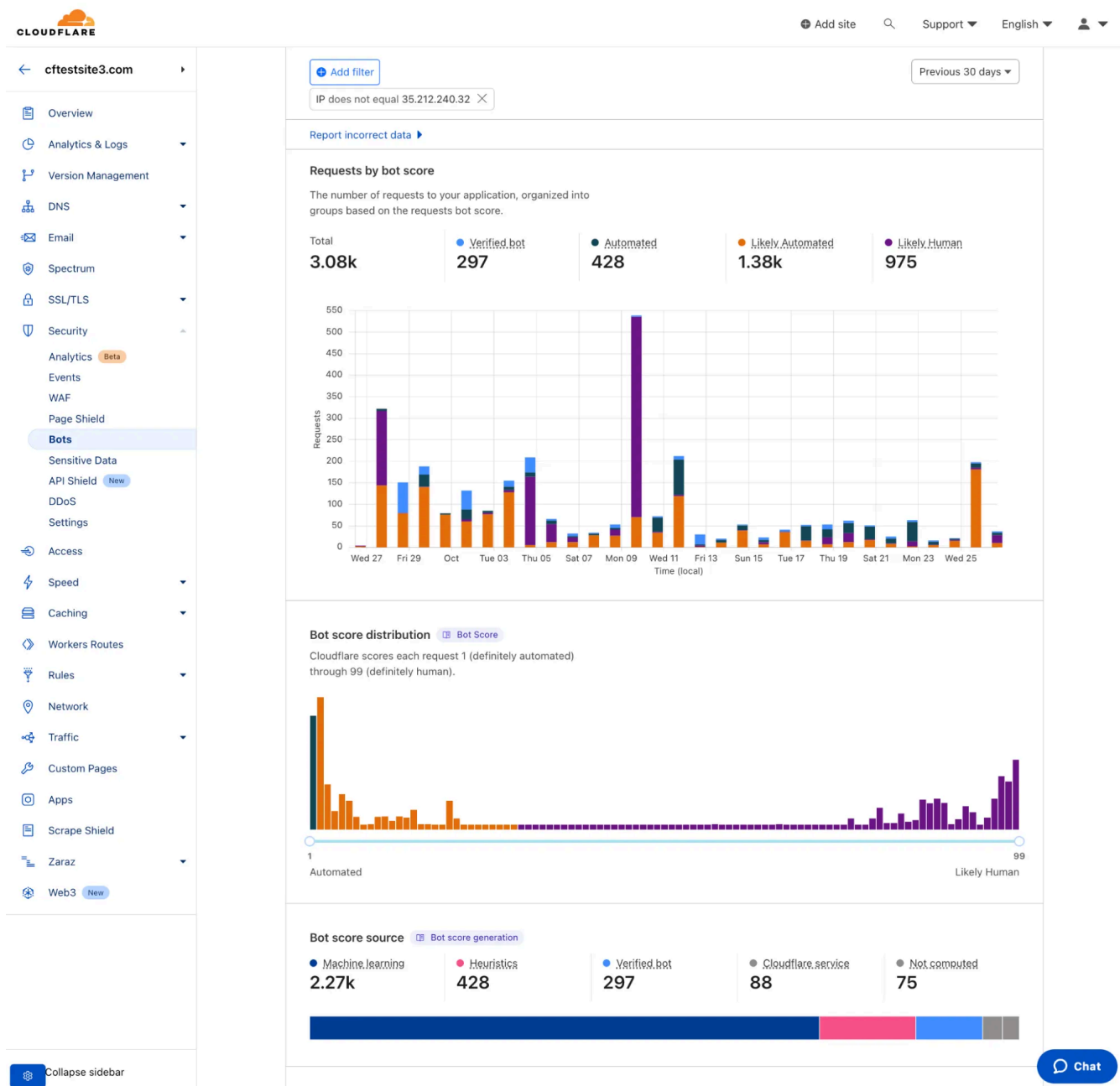


Figure 31 : Cloudflare Bot Management - Bot Analytics.

Summary

Cloudflare offers comprehensive application performance and security services. Customers can easily onboard and start using all performance and

security services by routing traffic to their origin server(s) through Cloudflare's network. Additionally, Cloudflare offers multiple connectivity options including Cloudflare Tunnel for securely connecting origin server(s) to Cloudflare's network.

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Last updated: Oct 9, 2025

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